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TRANSIENT ANALYSIS OF SERIES RLC CIRCUIT

AIM:

To determine transient response of a series RLC circuit using Multisim Simulator.

APPARATUS REQUIRED:

S.NO	COMPONENTS	SPECIFICATION	QUANTITY
1.	CRO	10MHz	1
2.	DCB, DIB	–	1
3.	Resistors	10 kΩ	
4.	Breadboard	–	1
5	Connecting wires	–	–
6.	PC with Multisim simulator		

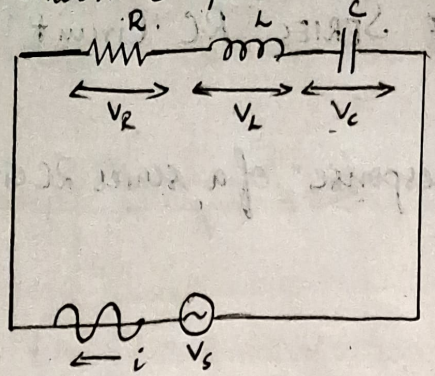
PROCEDURE:

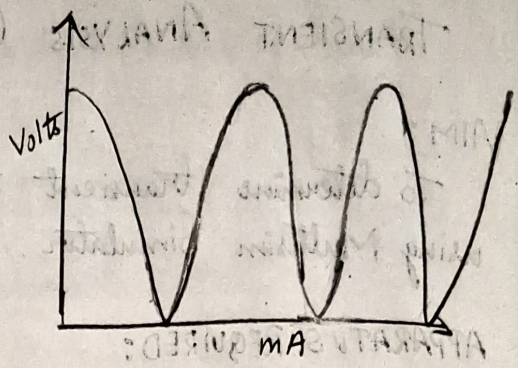
1. Make connection as per the circuit diagram.
2. Calculate the values $V_C(t)$, $V_R(t)$ $V_L(t)$ and write in table. Compare the calculated values and measured values of voltage in both the cases.

RESULT:

Thus the transient analysis of series RLC circuit is done practically using Multisim Simulator.

CIRCUIT DIAGRAM:





TABULATION:

Voltage across capacitor.
V_{in} Amplitude -

time(t)	$V_C(t)$ - charging cycle	$V_L(t)$ - Discharging cycle (V)
	observed value	calculated value
t=0	0	V
t=τ	0.632V	0.368V

Voltage across Inductor:

time (t)	$V_L(t)$ = Voltage across inductor
	observed value
t=0	10V
t=τ	3.68V

Voltage across Resistor:

Time (t)	$V_R(t)$ = Voltage across resistor
	observed value
t=0	0
t=τ	6.32